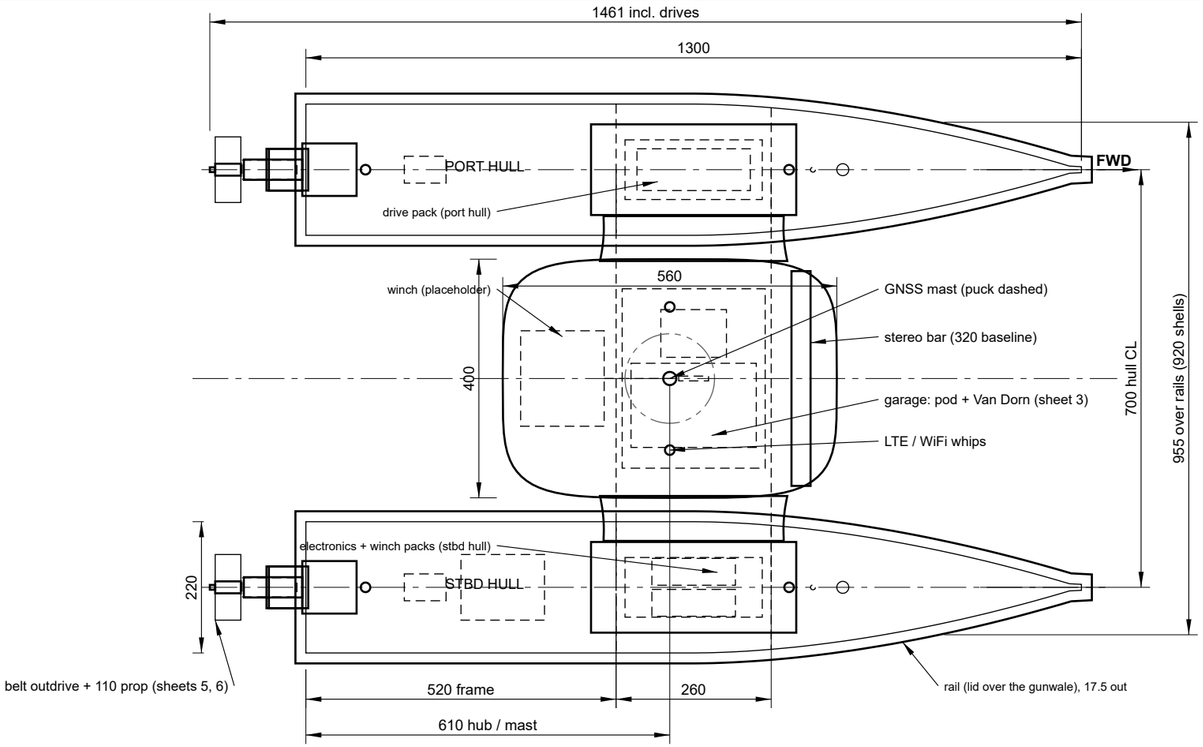
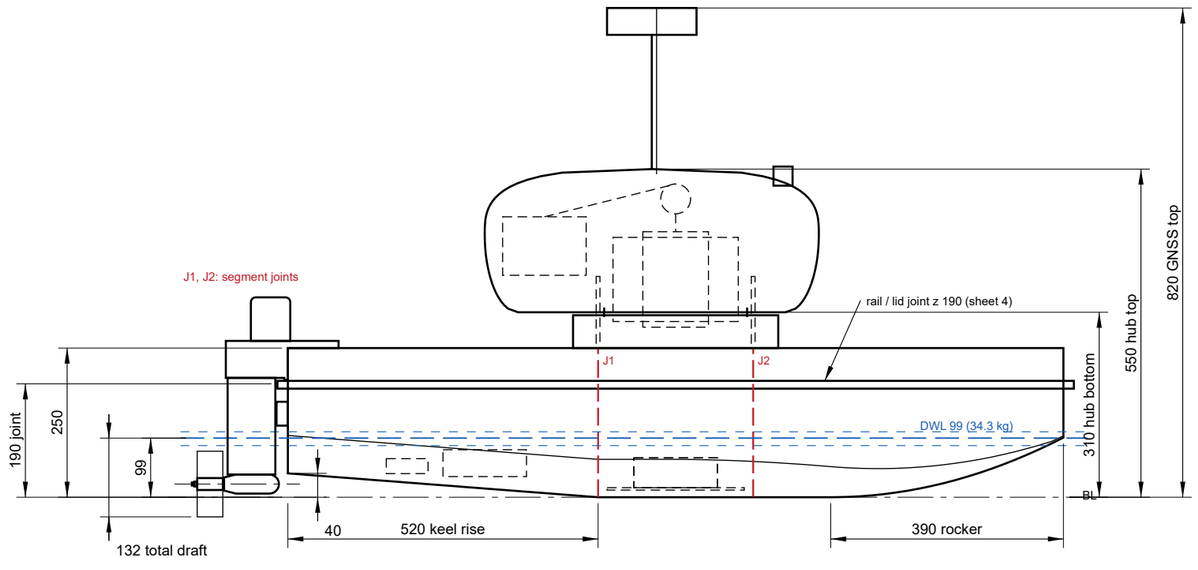


PLAN VIEW

SCALE 1:10



PROFILE (FROM STARBOARD)

SCALE 1:10

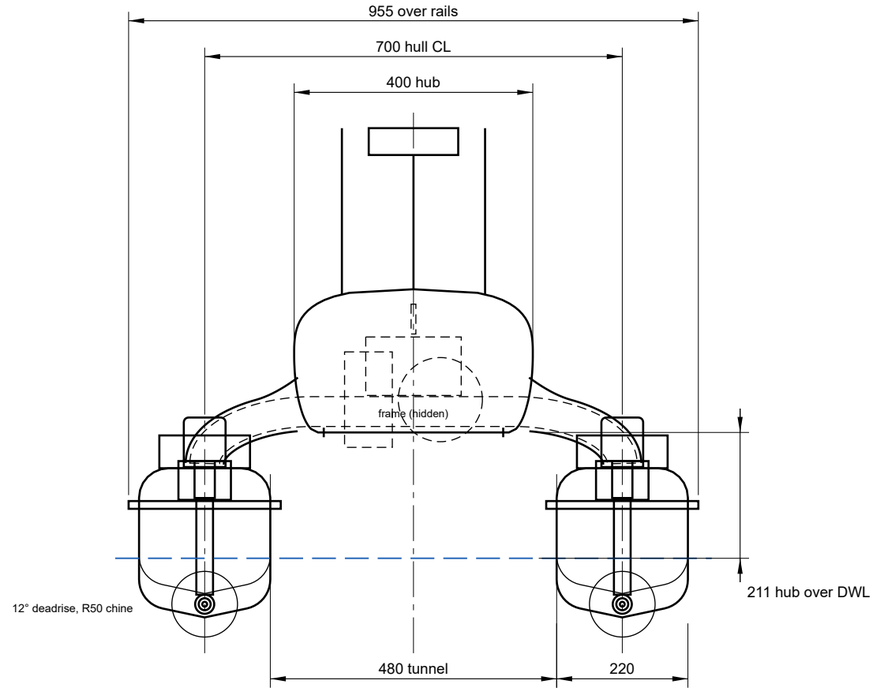
MODULES

Module	Design kg	Contents
Port tub	7.1	3 PLA segments + bulkheads, flange with 37 M4 inserts; drive pack, VESC, heat plate, leak probe, thrust pad
Stbd tub	6.6	the same shell; electronics + winch packs, Pi 5 + carrier, VESC, heat plate
Upper module	20.5	bridge (arch frames, arms, hub garage: winch, pod, Van Dorn; mast, cameras) + 2 lids + 2 belt outdrives
Boat	34.3	incl. 10% growth margin and a 2 L water sample; no robot arm

ASSEMBLY

1. Fit each tub's heat plate, packs and electronics, the leak probe (sheet 4) and the drive's thrust pad (sheet 6).
2. Bolt each foot to its lid from inside: 8x M6 x 60 up through the rib bosses, EPDM pad between.
3. Bolt each outdrive mount to its lid from inside: 4x M5 up through the aft deck, 2x through the aft wall.
4. Press the 3 mm cord into each tub's flange groove.
5. Hold the bridge + lids just above the tubs, plug the lid connectors inside, lower it; the drip lips locate.
6. Screw down the rails: 37 M4 per hull, 60 mm pitch within 150 mm of a frame, 100 elsewhere.

Waterlines: DWL 99 at 34.3 kg; thin blue = light 86 / max 110. Lid joint 91 above the DWL.
Wiring enters each hull only through IP68 bulkhead connectors in its lid. The packs sit low,
on aluminium plates on the hull bottoms, so the water carries their heat away.
Geometry comes from mechanical/hull_sizing.py; regenerate after changes.



VIEW FROM ASTERN

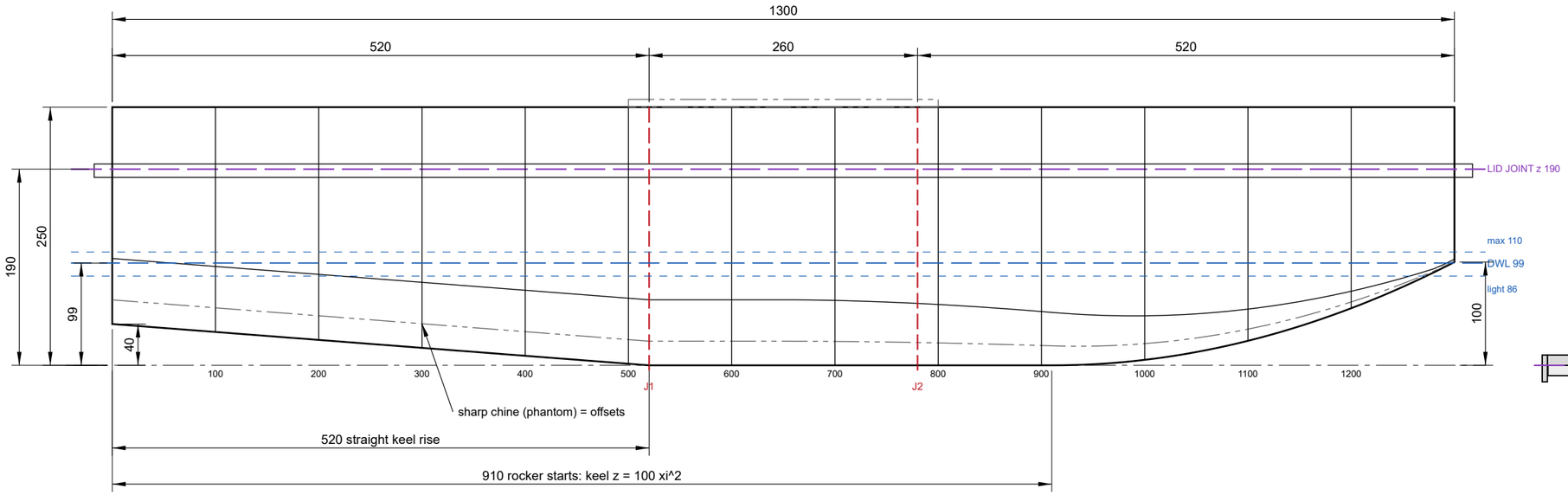
SCALE 1:10

D.R.E.A.M.S — autonomous survey catamaran

General arrangement: plan, profile, view from astern

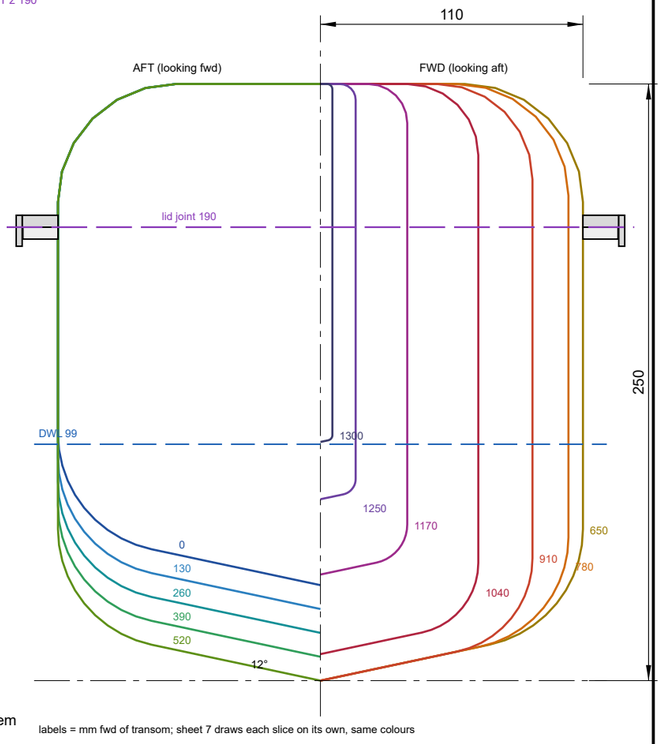
All dimensions mm unless noted. Third-angle projection. Do not scale the print; build from the numbers.

SCALE 1:10	SHEET 1 OF 7	REV D (hub garage, packs in the hulls, fixed drives)
DATE 2026-10-02	DRAWN Claude Code for M. Hong	SRC mechanical/drawings.py



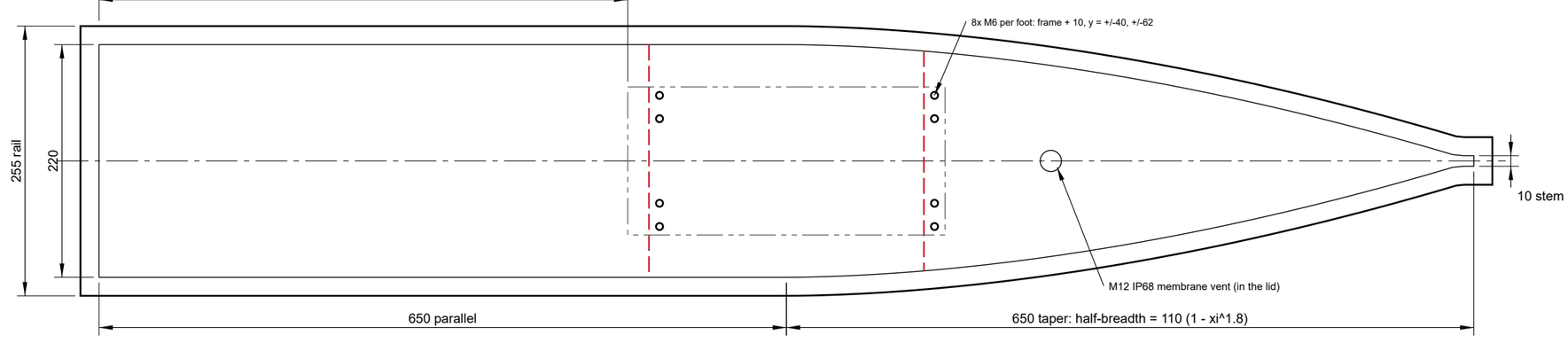
PROFILE (ONE HULL)

SCALE 1:5



BODY PLAN (with fillets)

SCALE 1:2.5



PLAN (ONE HULL)

SCALE 1:5

OFFSETS (sharp chine; mm from transom / baseline / hull centreline)

x	keel z	chine z	half-breadth	R chine / deck
0	40.0	63.4	110.0	50 / 50
130	30.0	53.4	110.0	50 / 50
260	20.0	43.4	110.0	50 / 50
390	10.0	33.4	110.0	50 / 50
520	0.0	23.4	110.0	50 / 50
650	0.0	23.4	110.0	50 / 50
780	0.0	22.1	103.9	47 / 47

x	keel z	chine z	half-breadth	R chine / deck
910	0.0	18.9	88.9	40 / 40
1000	5.3	21.0	73.9	33 / 33
1100	23.7	35.1	53.3	24 / 24
1170	44.4	52.2	36.4	16 / 16
1250	76.0	79.1	14.8	7 / 7
1300	100.0	101.1	5.0	2 / 2

CONSTRUCTION (open tub + removable lid)

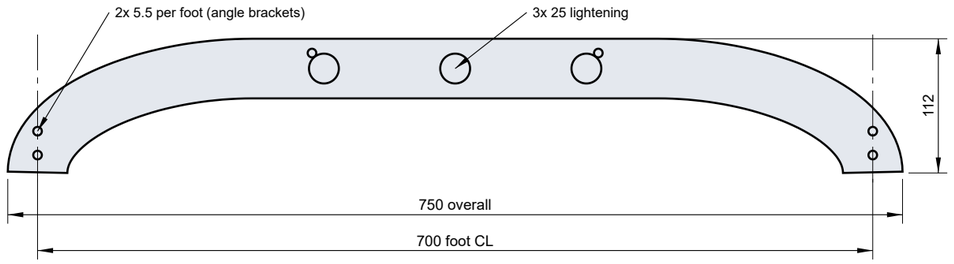
- Loft the sharp-chine hull from the offsets, fillet chine R50 and deck edge R50 (0.45 x half-breadth near the stem), split at the lid joint plane z 190.
- PLA on the Elegoo OrangeStorm Giga: 0.6-0.8 nozzle, 4 walls (2.4), +3-5 % flow. Glass outside (200 g/m2; 2nd layer on keel, chines, stem), 2 coats epoxy inside.
- Tub: 3 segments (520 / 260 / 520) split at the frames, each printed standing on its aft 2.4 bulkhead (the first on the transom).
Bulkheads stop 5 under the joint plane, with limber holes at their feet so a leak runs to the probe.
- Lid: 3 segments on end; lid segments 2 and 3 stand on their 8 mm ribs (J1 / J2) with the foot-bolt bosses.
- Joints J1, J2: 8 mm half-lap spigots, bonded with thickened epoxy (sheet 3). Epoxy only, slow hardener: PLA softens ~55 C.
- Rail: 15 mm flanges, 3 mm cord in a 5 x 2.2 groove, 37x M4 into heat-set inserts (sheet 4); epoxy and sand the flange: layer lines cross the gasket.
- One sealed space per hull: leak probe at the keel low point (carrier J8 port / J9 stbd), M12 vent in the lid.
- Shell 4.6 kg per hull incl. bulkheads, ribs, stiffeners, glass, epoxy, + 1.0 kg rail. Pressure-test (<= 0.03 bar) before fitting out.

D.R.E.A.M.S — autonomous survey catamaran

Demi-hull lines, lid joint plane, print segments, offsets and fillets

All dimensions mm unless noted. Third-angle projection. Do not scale the print; build from the numbers.

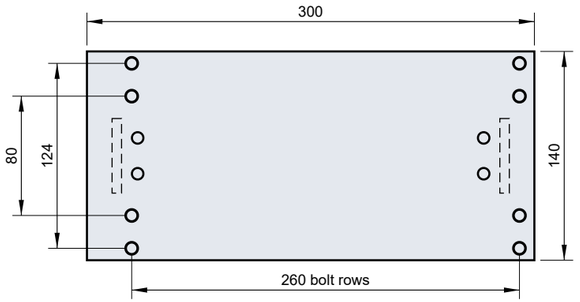
SCALE 1:5 / 1:2.5	SHEET 2 OF 7	REV D (hub garage, packs in the hulls, fixed drives)
DATE 2026-10-02	DRAWN Claude Code for M. Hong	SRC mechanical/drawings.py



1/4 in (6.35) 6061-T6, 2 req. Section 50 deep, +/-25 from the arch centreline; feet stand on the foot plate, 7.5 above the lid deck. Cut file: cad/arch_frame_plate.dxf

ARCH FRAME PLATE (FROM AHEAD)

SCALE 1:5



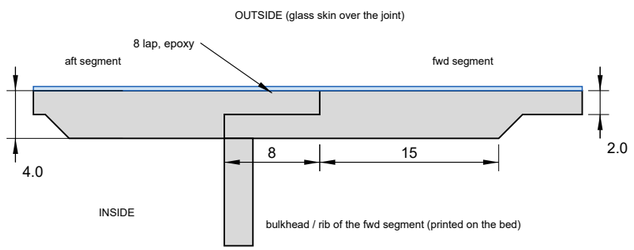
Arch centreline (stbd arm)	y	z
hub span, straight	0 to 170	345
Bezier P0 (arm root)	170	345
Bezier P1	260	345
Bezier P2	350	308
Bezier P3 (foot)	350	257.5

Port arm is the mirror image.
Tangent horizontal at P0, vertical at P3.

6 mm 6061. Frames (hidden) at -130 / +130. 8x M6 tapped through at frame + 10 (bolts come up from inside the lid); 2x 5.5 per angle bracket

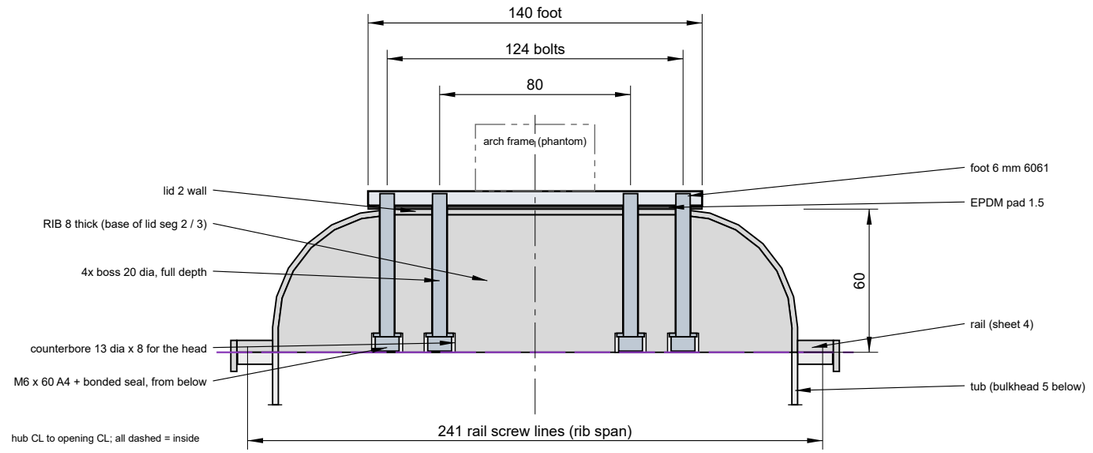
FOOT PLATE (PLAN, 2 req)

SCALE 1:4

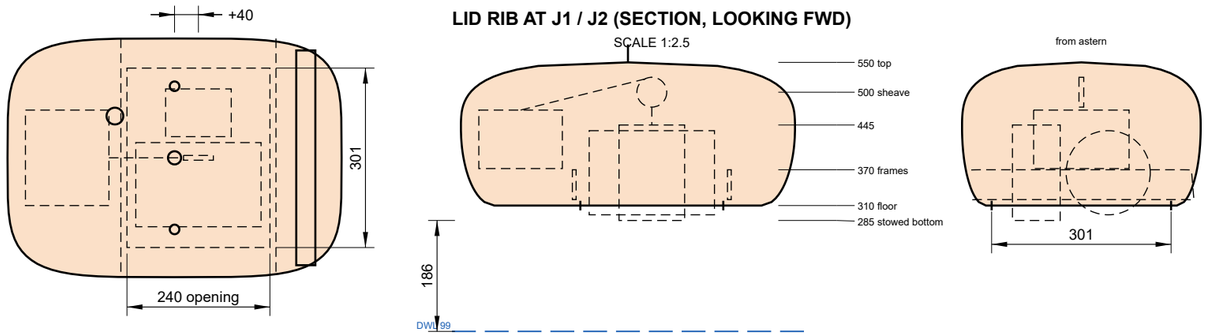


SEGMENT JOINT, TUB + LID (45° chamfers print unsupported)

SCALE 2:1



LID RIB AT J1 / J2 (SECTION, LOOKING FWD)



HUB GARAGE: WINCH, POD, VAN DORN (PLAN, SIDE, FROM ASTERN)

SCALE 1:10

Pod 110 x 80 x 160 (port) and a 2 L horizontal Van Dorn, 5 in PVC, 141 x 210 (stbd), side by side under the sheave: margins +30 / +70 mm through the opening. Winch, pod and sampler are placeholders until they're sized; the winch sits on a bracket off the aft frame, the sheave over the opening on the garage holes in the frames.

HARDWARE (whole boat)

Qty	Item
2	Arch frame plate, 1/4 in 6061-T6, laser / waterjet cut (DXF)
2	Foot plate, 6 mm 6061, 300 x 140, 8x M6 tapped
4	Angle bracket 25 x 25 x 3 Al, 40 long (frame to foot)
16	M6 x 60 A4 socket cap + M6 bonded seal washer (feet, from inside)
2	EPDM pad 1.5 mm, 300 x 140 (under each foot)
74	M4 x 12 A4 pan head + washer, and M4 x 8.1 brass heat-set insert (rails)
2	3 mm EPDM / silicone cord, 2.85 m (one loop per tub)
1	Hub cowl, 2 arm fairings, 2 foot shoes (printed PLA, painted light)
1	Garage: winch bracket off the aft frame, sheave bracket, printed docking cone
2	Belt outdrive, complete (sheets 5, 6)
2	M12 IP68 membrane vent (one per lid)
4	IP68 bulkhead connector in the lids: motor (x 100) and bridge (x 810), per hull
2	Leak probe at each tub's keel low point (carrier J8 port / J9 stbd)

LOAD PATH

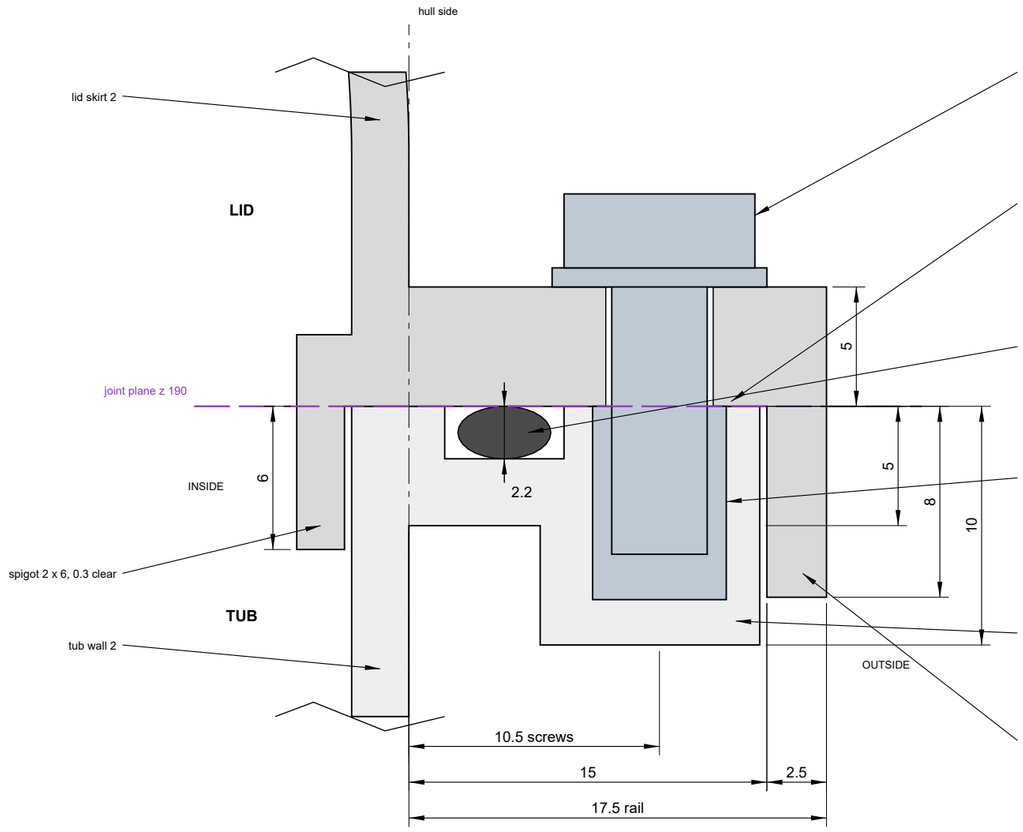
Each frame is a portal fixed at its feet.
Worst case: hulls pitching out of phase, a_cg 2 g
147 N m per frame -> 56 MPa,
SF 5.0 on 6061-T6 yield.
Foot: pivots on its far edge; the 8
M6 in the rib bosses hold it: SF 3.2.
Lid rib carries it to the rail: SF 4.3.
Rail: 3 M4 inserts per frame: SF 4.4.
Nothing goes through a tub: all
hardware bolts to the lids.
Details: docs/hull_and_crossbeam_sizing.md

D.R.E.A.M.S — autonomous survey catamaran

Arch frame, foot plate, lid rib, hub garage, segment joint

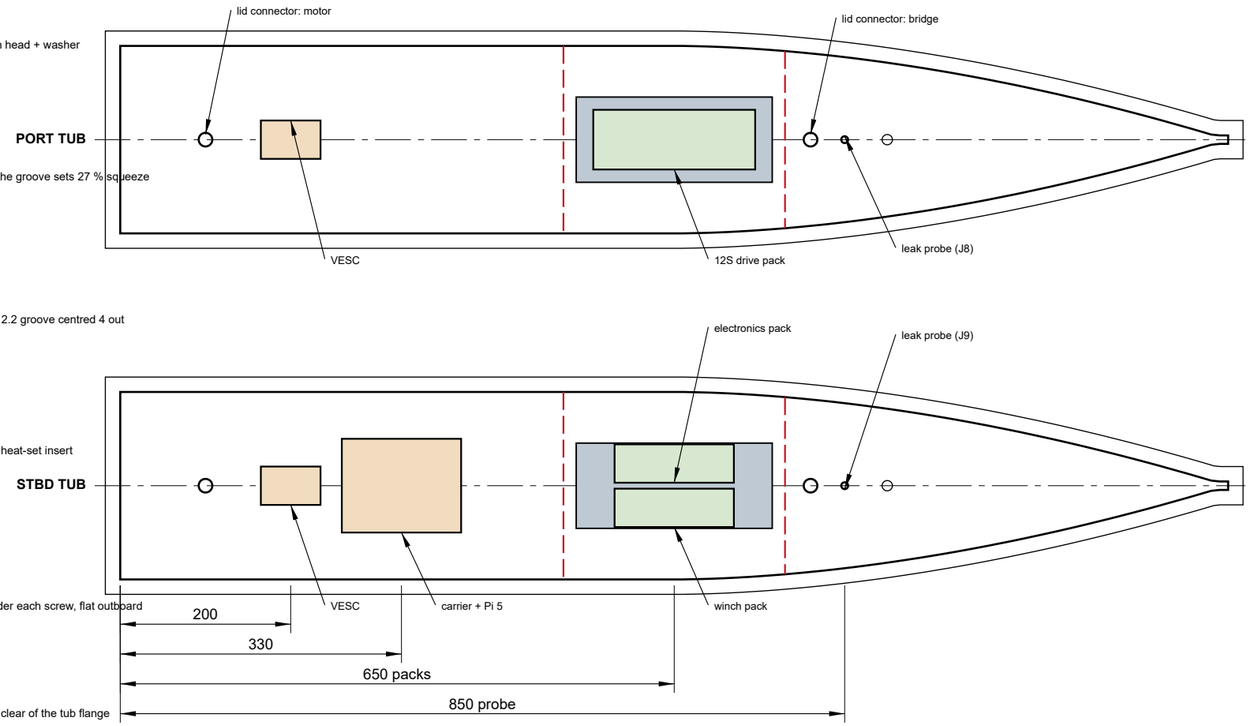
All dimensions mm unless noted. Third-angle projection. Do not scale the print; build from the numbers.

SCALE as noted	SHEET 3 OF 7	REV D (hub garage, packs in the hulls, fixed drives)
DATE 2026-10-02	DRAWN Claude Code for M. Hong	SRC mechanical/drawings.py



RAIL: LID OVER THE GUNWALE (SECTION THROUGH A SCREW)

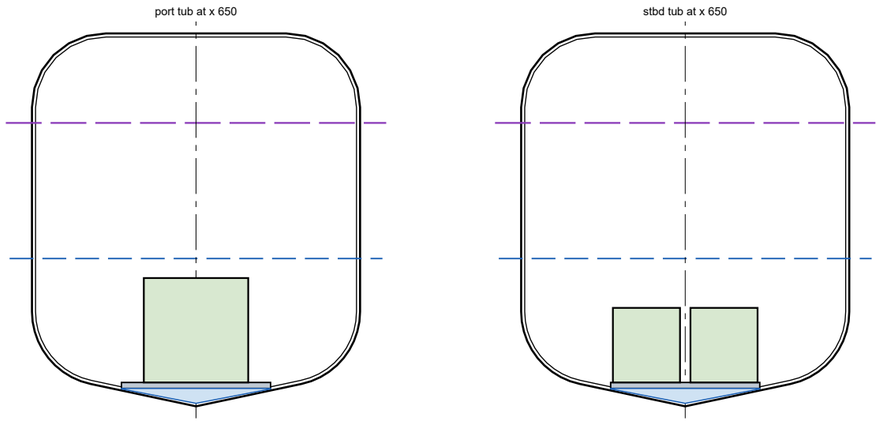
SCALE 4:1



INSIDE THE TUBS (PLAN, PROPOSED)

SCALE 1:7

SECTIONS AT x 650, looking fwd, 1:4: the heat plate is bedded on thickened epoxy (blue) on the V bottom, the packs sit on the plate. DWL dashed blue, lid joint purple.



WIRING (everything enters a hull through its lid connectors)

- 12S: drive pack (port) -> port VESC; to the stbd VESC across the bridge (port arm, hub, stbd arm).
- CAN: carrier (stbd) -> stbd VESC -> across the bridge -> port VESC (node IDs 1, 2).
- Electronics pack (stbd) -> carrier -> Pi 5 on the carrier's 5 V. Never also power the Pi from USB-C.
- Winch pack (stbd) -> carrier's fail-safe winch kill -> up the stbd arm -> winch in the hub garage.
- Pod tether (RS-485 + pack voltage) from carrier J2 -> up the stbd arm -> the winch's slip ring.
- Mast (GNSS, LTE, WiFi), stereo cameras, e-stop -> down the stbd arm. Leak probes: J8 port, J9 stbd.
- Motor wires: outdrive -> lid connector at x 100 -> that hull's VESC.

HEAT AND WATER

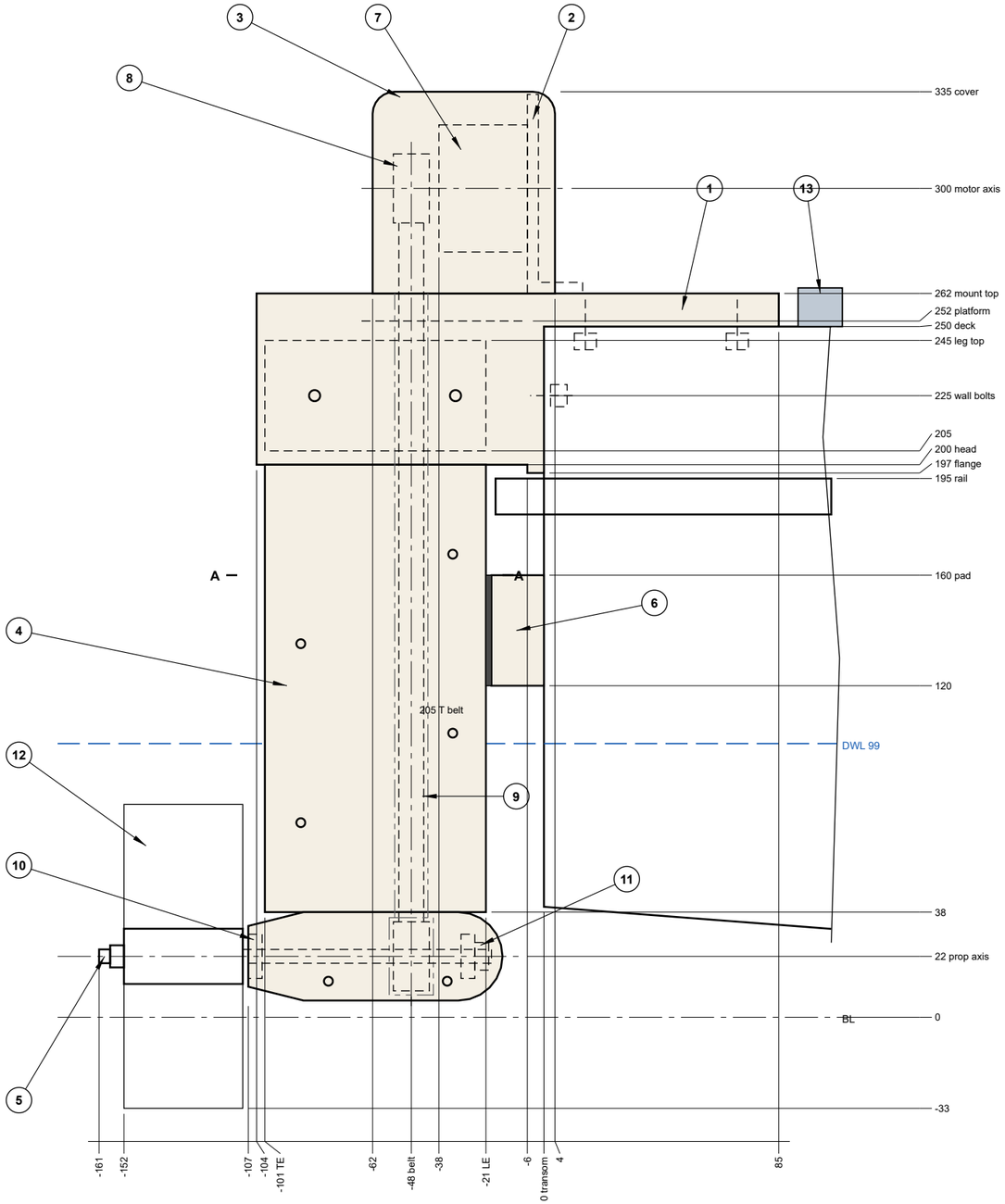
- VESCs and packs sit on 6061 plates bedded on the hull bottom, so the water takes the heat; PLA softens ~55 C. Log the hull air temperature (the carrier's SHT45s).
- Limber holes at the bulkhead feet send any leak to the probe at x 850, forward of the packs.
- Lifting a lid: unscrew the rail, raise it ~100 mm on the connector slack, unplug inside, lift away.

D.R.E.A.M.S — autonomous survey catamaran

Hull lid joint (rail) and what goes inside each hull

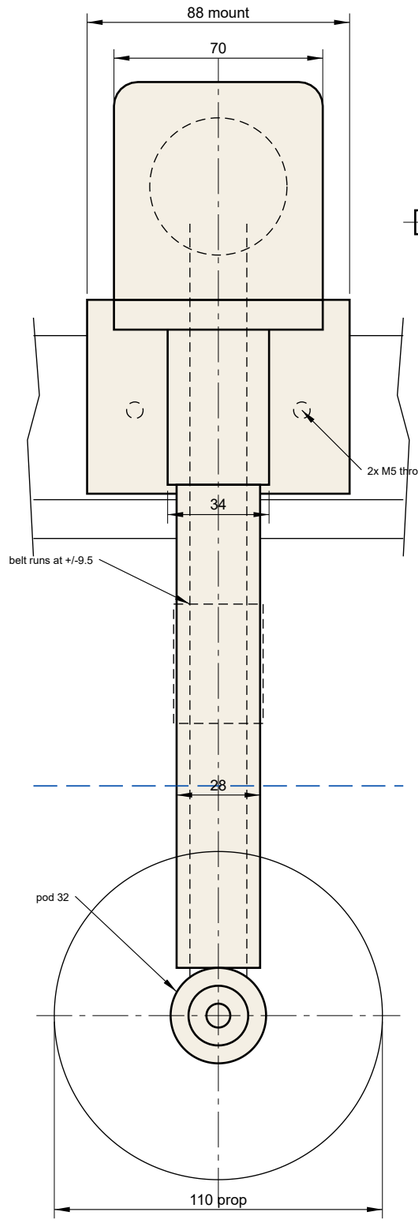
All dimensions mm unless noted. Third-angle projection. Do not scale the print; build from the numbers.

SCALE as noted	SHEET 4 OF 7	REV D (hub garage, packs in the hulls, fixed drives)
DATE 2026-10-02	DRAWN Claude Code for M. Hong	SRC mechanical/drawings.py



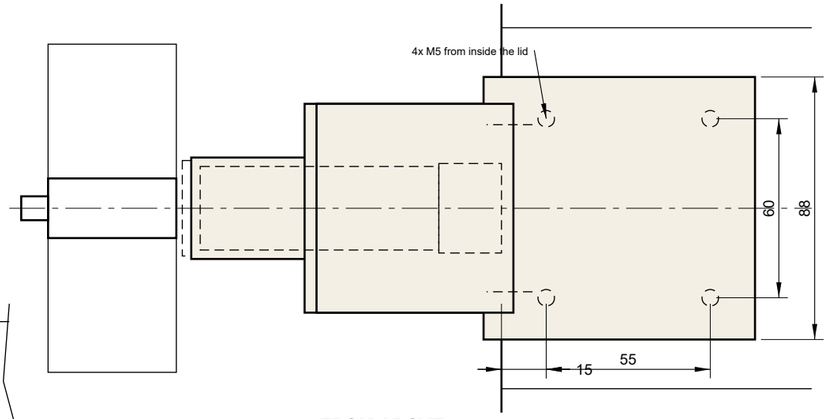
SIDE VIEW, STARBOARD DRIVE (FIXED)

SCALE 1:2



FROM ASTERN

SCALE 1:2



FROM ABOVE

SCALE 1:2

KEY POINTS (mm; x fwd of the transom, z above the baseline, y off the hull CL)

Feature	x	z	y
Mount: deck flange	0 to 85	250 to 262	+/-44
Mount: flange down the lid's aft wall	-6 to 0	197 to 262	+/-44
Mount: head block	-104 to -6	200 to 262	+/-17
Mount: platform (under the cover)	-66 to -6	252 to 262	+/-35
Mount: leg socket	-101 to -21	205 to 245	+/-14
Deck bolts, 4x M5 up from inside	15, 70	250	+/-30
Wall bolts, 2x M5 fwd from inside	10 to -6	225	+/-28
Thrust pad (+ 2 EPDM face)	-21 to 0	120 to 160	+/-15
Motor plate, 4 mm 6061	-6 to -2	262 to 334	+/-25
Motor can (41xx)	-38 to -6	axis 300, dia 46	0
Motor cover	-62 to 4	262 to 335	+/-35
Belt plane (both pulleys)	-48	300 / 22	runs +/-9.5
Leg, NACA 0035, 80 x 28	TE -101, LE -21	38 to 245	+/-14
Lower pod, dia 32	tail -107, nose -15	axis 22	0
Bearings 625-2RS	-30 to -25; -107 to -102	axis 22	
Shaft collar / shaft, dia 5	-25 to -20 / -161 to -19	axis 22	
Prop hub 20 / disc 110 / nylock	-152 to -109 / -157	22 +/- 55	
Motor-wire connector (lid deck)	100	250	0

PARTS (the balloons; 1-6 are drawn on sheet 6, 7-13 are bought)

1. Mount
2. Motor plate
3. Motor cover
4. Leg + pod halves
5. Prop shaft
6. Thrust pad
7. Motor
8. Pulleys x2
9. Belt
10. Bearings x2
11. Shaft collar
12. Prop, washer, nylock
13. Lid connector

HOW TO USE THIS SHEET

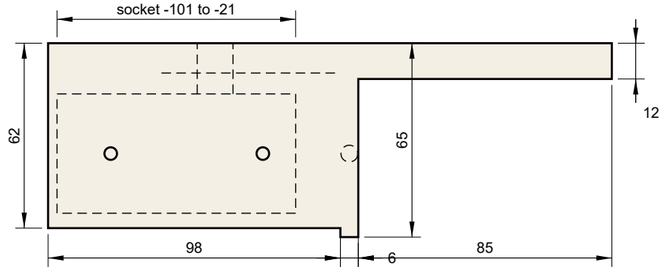
- Datums: x from the transom (the lid's aft face), z from the baseline, y from the hull centreline. Model the drive in place in the boat assembly.
- Fixed: no pivot. The leg runs 3.5 mm clear of the rail (as close as the rail allows); its nose rests on the thrust pad on the transom. Tightest gap: wall flange over rail 2.0 mm.
- At full thrust (55 N) the pad takes 131 N and the mount 76 N; the leg bends 6.5 N m at the pad instead of 11.2 N m at the mount. Reverse hangs on the bolts.
- Belt tension: slide the motor up its slotted plate (+/-4). The port drive is the mirror image with an LH prop. Parts, sizes and the motor note: sheet 6.

D.R.E.A.M.S — autonomous survey catamaran

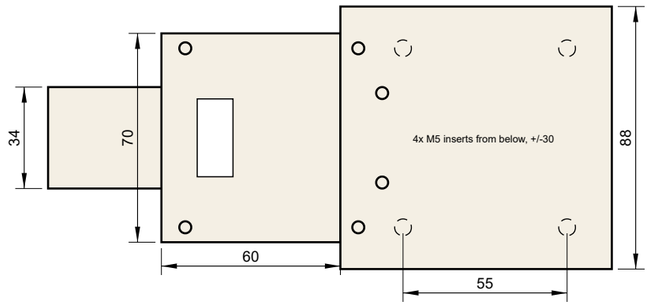
Belt outdrive (fixed), assembled: side, from astern, from above, key points (for CAD)

All dimensions mm unless noted. Third-angle projection. Do not scale the print; build from the numbers.

SCALE 1:2	SHEET 5 OF 7	REV D (hub garage, packs in the hulls, fixed drives)
DATE 2026-10-02	DRAWN Claude Code for M. Hong	SRC mechanical/drawings.py

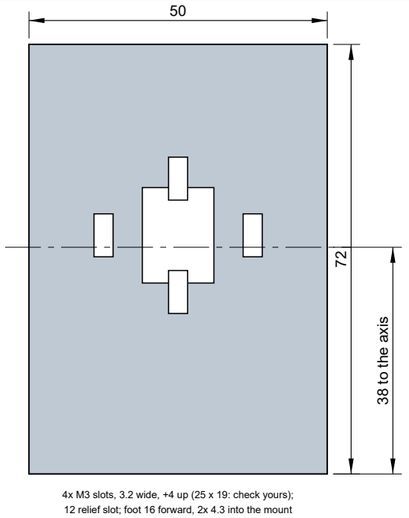


socket = leg section + 0.2, 40 deep, 2x M4 across; wall flange: 2x M5 inserts



1 MOUNT (side, above)

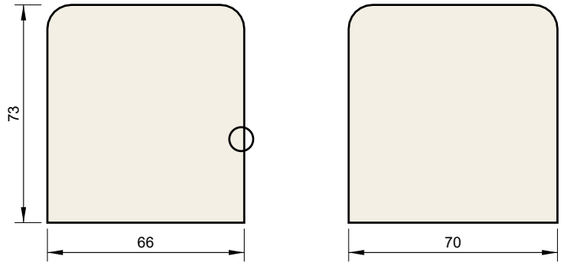
SCALE 1:2



4x M3 slots, 3.2 wide, +4 up (25 x 19: check yours);
12 relief slot; foot 16 forward, 2x 4.3 into the mount

2 MOTOR PLATE (4 mm 6061, L)

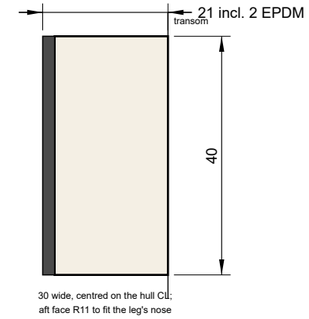
SCALE 1:1



2 wall, R8 edges, open at the bottom; 4x M3 tabs into the mount top; 8 grommet in the front wall for the motor wires; 10 x 3 drain slot at the front.

3 MOTOR COVER (side, astern)

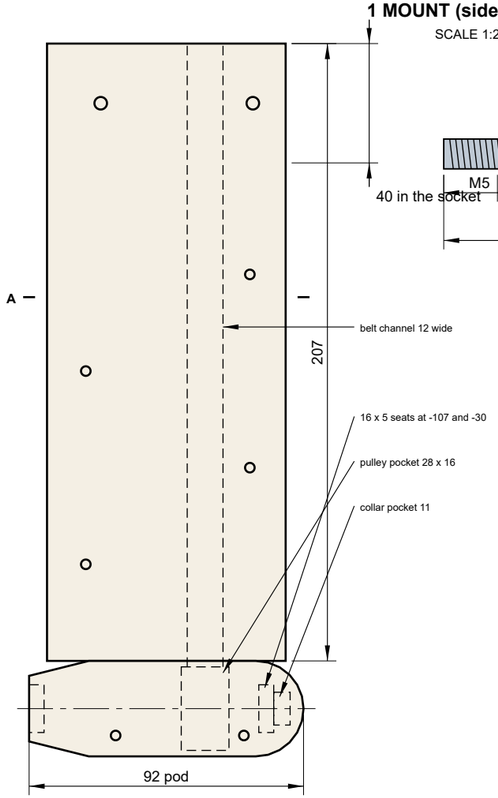
SCALE 1:2



30 wide, centred on the hull CL;
aft face R11 to fit the leg's nose

6 THRUST PAD

SCALE 1:1

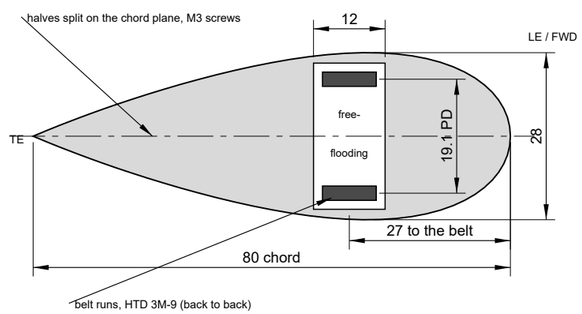


4 LEG + POD HALF (stbd half; split on y = 0)

SCALE 1:2

5 PROP SHAFT (316, ends at x -161 and -19)

SCALE 1:1



SECTION A-A THROUGH THE LEG

SCALE 1:1

PURCHASED PARTS (per drive; 2 drives)

No.	Qty	Item
7	1	Outrunner, 41xx (46 x 32 can), 25 A+. KV: see the note
8	2	HTD 3M pulley, 20T (19.1 PD), 9 mm belt; bores to the motor / 5 mm shaft
9	1	HTD 3M belt, 9 mm, 205 T (615 mm), PU with aramid or stainless cords
10	2	625-2RS bearing, stainless (5 x 16 x 5): they run wet
11	1	Shaft collar, 5 mm bore, 10 OD x 5, set screw
12	1	Prop 110 (props/toroidal_..._RH_shaft stbd, _LH_ port) + washer + M5 nylock A4
13	1	IP68 bulkhead connector, 3-pin, in the lid's aft deck at x 100
-		6x M5 x 20 A4 + washers (4 up through the deck, 2 through the aft wall), 3 mm Al backing plates
-		6x M5 heat-set inserts in the mount; 2x M4 x 40 A4 + nylocks across the socket
-		~12x M3 A4 + inserts (leg/pod halves, cover); 2x M4 x 8 (motor plate foot)

NOTES

- Print 1, 3 and 4 in ABS or ASA (the EXP S5 lists ABS), not PLA: the motor and the sun heat them. 6 walls or 100 % infill round the socket and the inserts. The mount can be milled from 6061 later.
- Bond 6 to the transom with thickened epoxy; it carries the forward thrust. The leg lifts off with the lid; nothing on the drive bolts to the tub.
- Loads at 55 N forward: pad 131 N, mount 76 N, leg 6.5 N m at the pad (11.2 N m with no pad).
- MOTOR KV. V3's drive rail is 12S (~44 V). Rev C picked 380 KV for an 18 V pack; on 12S it would run at ~10-25 % duty from cruise to top speed. ~150-200 KV suits 12S. Most motors with enough power at that KV are a size up (50xx, ~55 dia): that fits the 70 cover with the motor axis ~5 mm up. The drawings keep the 41xx envelope until you pick one.

MODELING ORDER (SolidWorks)

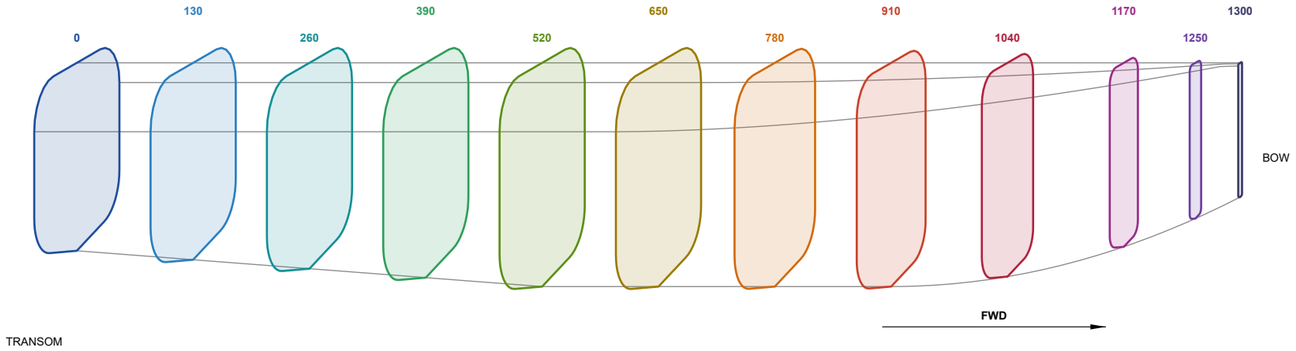
1. Drive assembly layout sketch on the Front plane: transom (x 0), deck (z 250), belt plane, motor and prop axes. Origin = transom / baseline / hull CL, like the hull.
2. Mount: extrude the side profile +/-17 (head block); add the deck and wall flanges +/-44 and the platform +/-35; cut the socket (leg section + 0.2), the belt slot and the holes.
3. Leg + pod: section A-A at z 38 extruded to 245; pod revolved on the prop axis; combine, cut the pockets, split on y = 0. Mirror the half for the other side.
4. Pad, plate, cover; then mate everything fixed and run Interference Detection.

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Belt outdrive (fixed), part by part (for CAD), purchased parts and modeling order

All dimensions mm unless noted. Third-angle projection. Do not scale the print; build from the numbers.

SCALE as noted	SHEET 6 OF 7	REV D (hub garage, packs in the hulls, fixed drives)
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WHERE EACH SLICE IS (ONE HULL)

SCALE sketch, not to scale

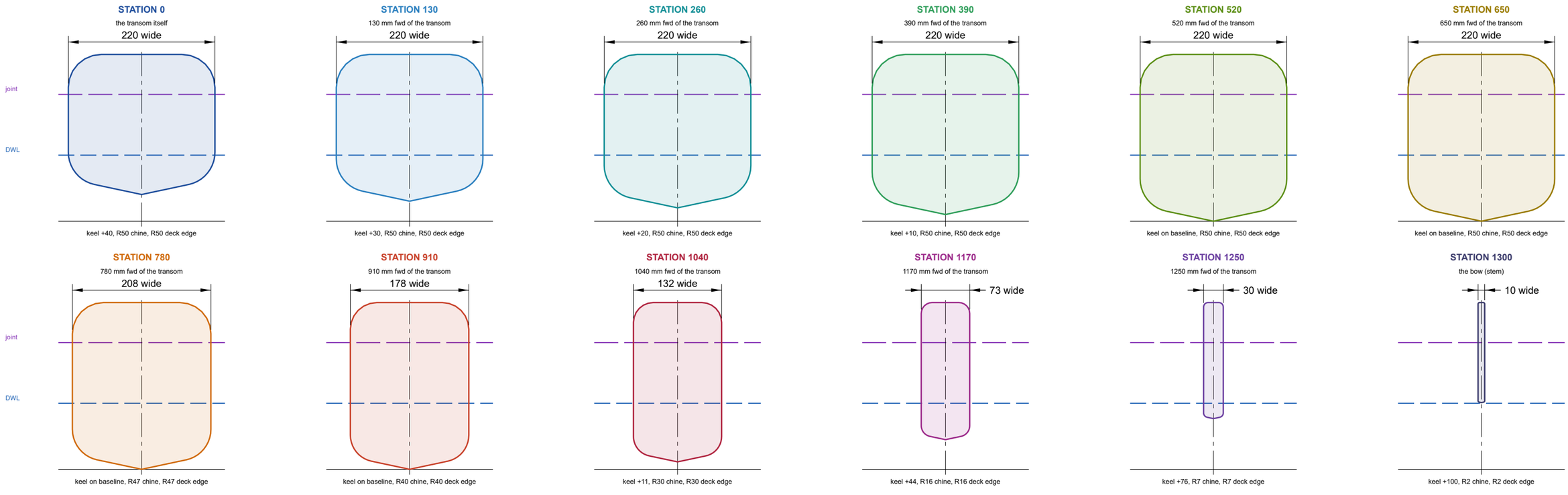
HOW TO READ THE BODY PLAN (SHEET 2)

Each coloured shape is one slice straight across ONE hull, like a slice of bread. Its number is where the slice is: mm forward of the transom.

Sheet 2 stacks all the slices on one drawing. The hull is the same on both sides, so it draws only half of each slice: the stern half of the boat on the left of the centreline, the bow half on the right, in these colours.

Reading from stern to bow:
- 0 to 520: the same shape, but the keel lifts toward the transom (+40 mm at 0), so the stern draws less water.
- 520 to 650: the widest part (220 mm), keel on the baseline.
- 650 to 1300: it narrows toward the bow; from 910 the keel also rises, to +100 at the bow, which ends in a 10 mm flat face.

In each box: blue dashes = waterline (99), purple = lid joint (190).
In SolidWorks, each box is one sketch on a plane at that x (guide, A2).



Scale 1:6. Full slices, both halves, looking forward from the stern.

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Hull sections one at a time: a reading aid for the body plan

All dimensions mm unless noted. Third-angle projection. Do not scale the print; build from the numbers.

SCALE 1:6 / sketch	SHEET 7 OF 7	REV D (hub garage, packs in the hulls, fixed drives)
DATE 2026-10-02	DRAWN Claude Code for M. Hong	SRC mechanical/drawings.py